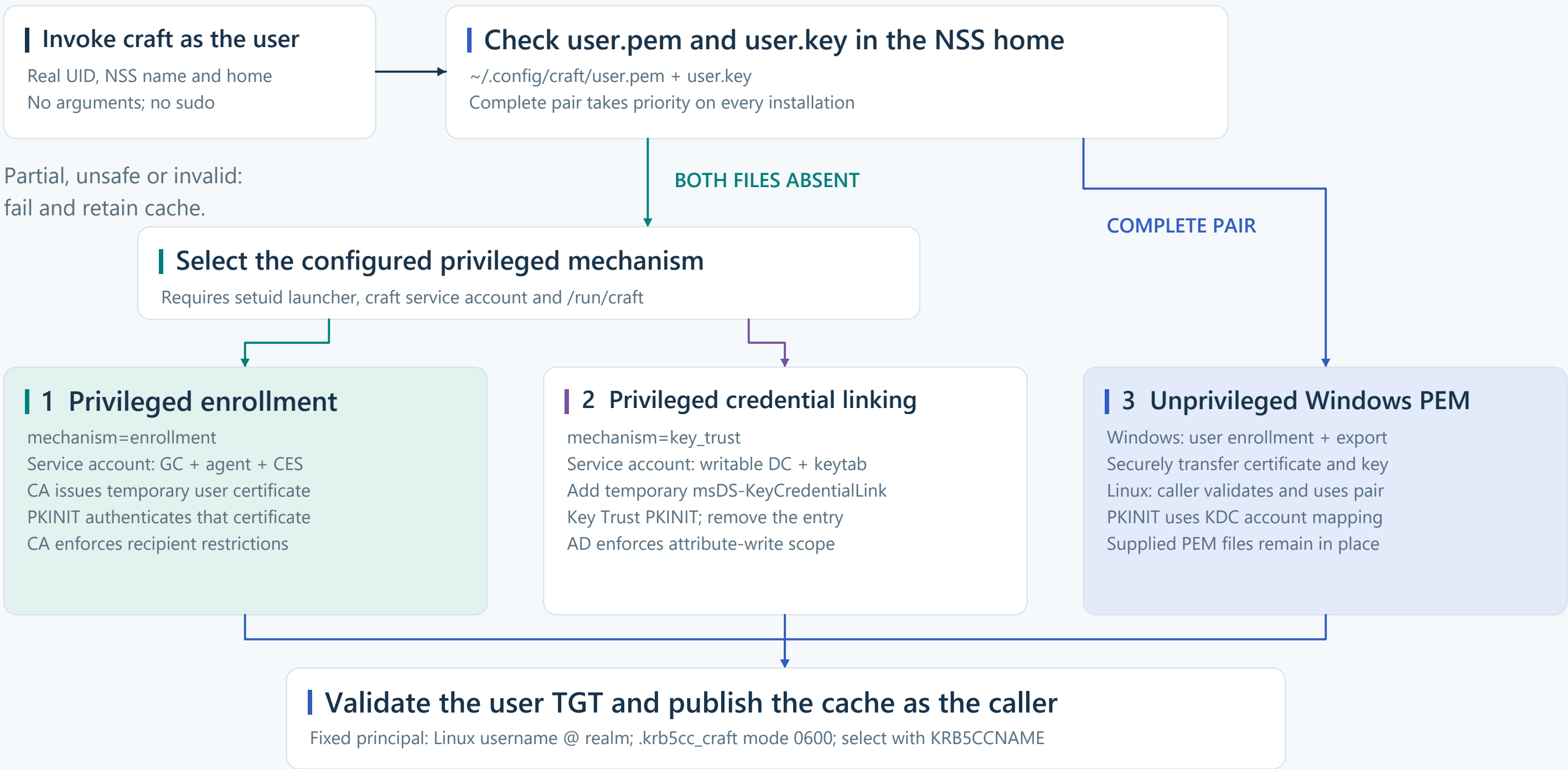


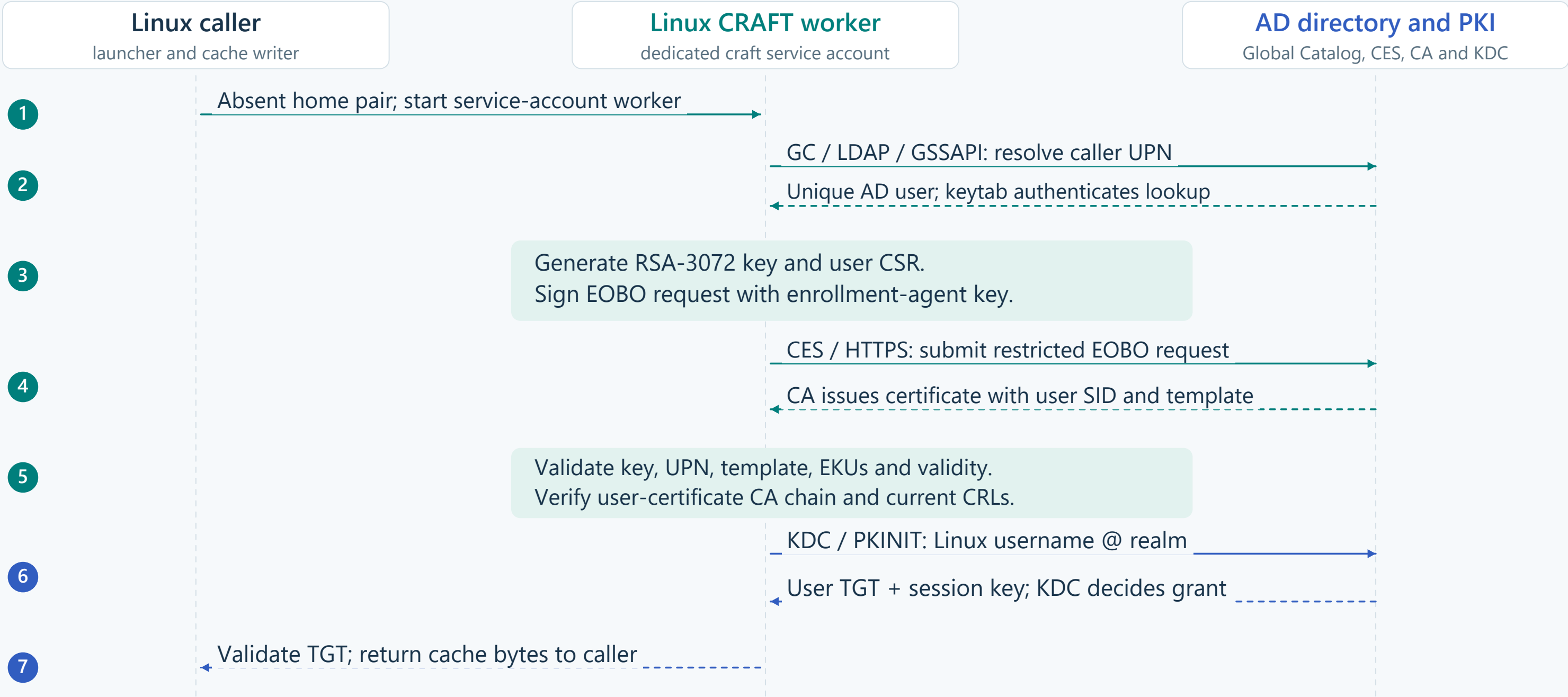
Three Linux operating modes

A supplied PEM pair takes priority. Only an absent pair selects the configured privileged mechanism.



Mode 1 Privileged certificate enrollment

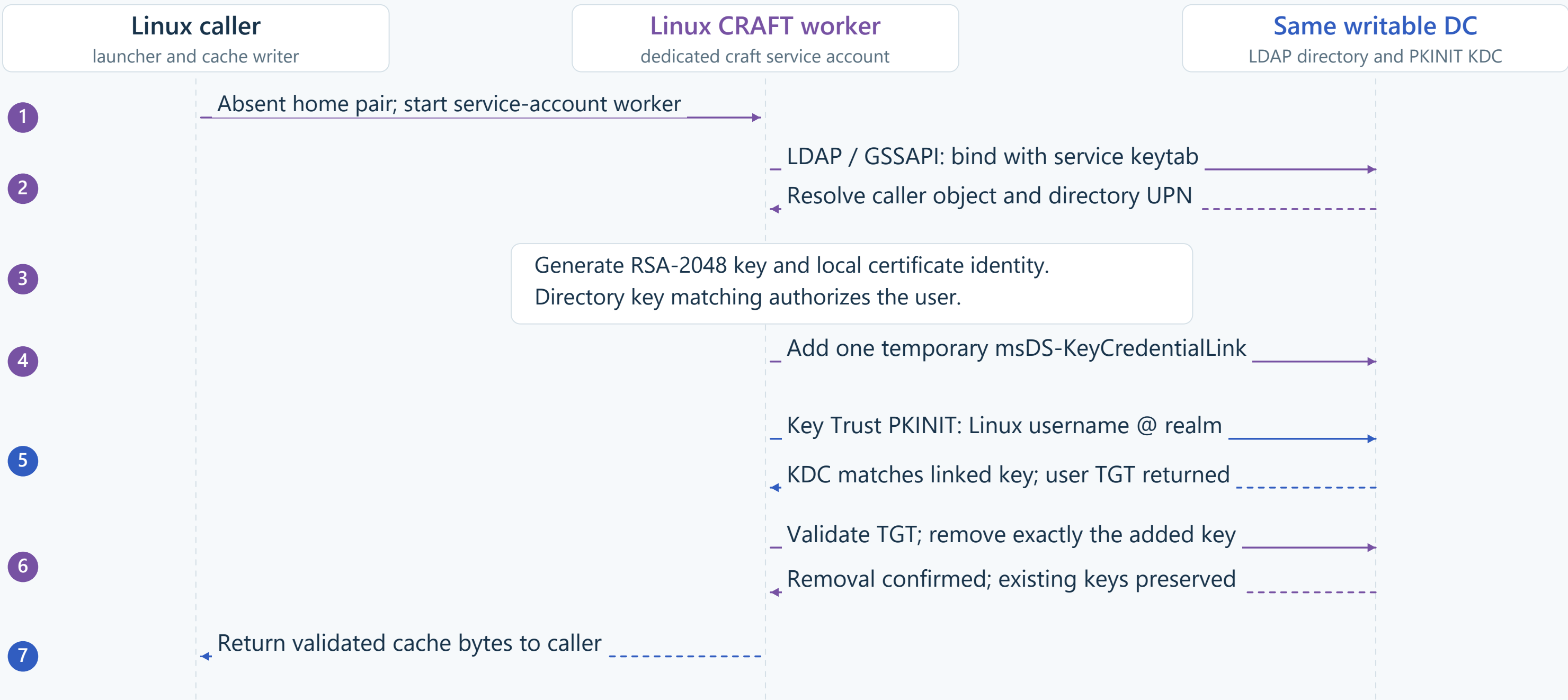
Linux orchestrates CA enrollment first, then PKINIT. Select mechanism=enrollment with both home PEM files absent.



Caller publishes .krb5cc_craft mode 0600. Temporary user key and certificate are released.

Mode 2 Privileged credential linking

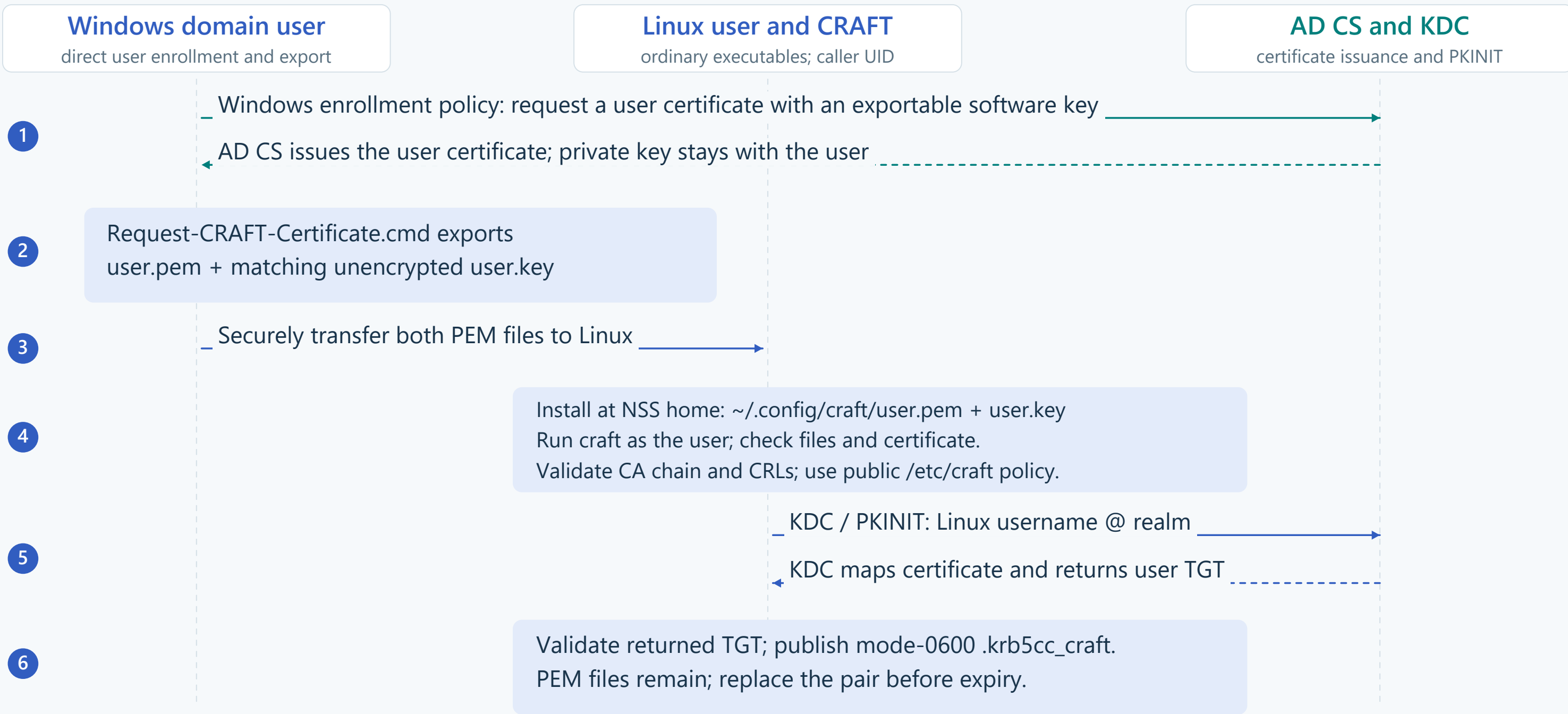
Linux links a temporary AD public key, performs Key Trust PKINIT, and removes the entry before publishing the TGT.



Caller publishes .krb5cc_craft mode 0600. Cleanup also runs after acquisition failure.
Failed cleanup blocks publication and reports a residual key for administrator removal.

Mode 3 Unprivileged Windows-exported certificate

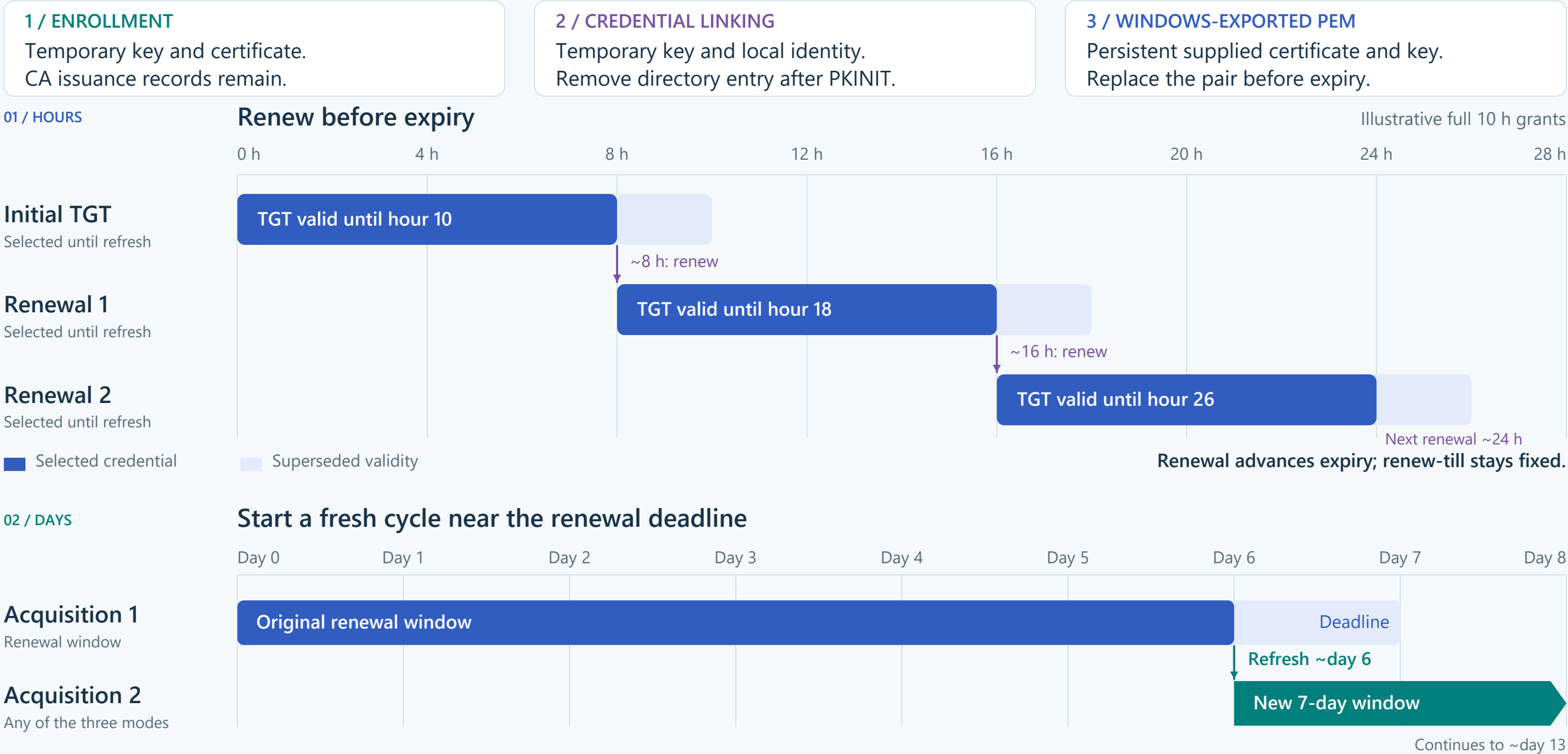
Windows enrolls and exports the certificate. Linux uses the transferred PEM pair entirely as the caller.



No Linux `setuid` bit, service account, directory keytab, LDAP lookup, CES or `/run/craft`.

Ticket clocks and certificate storage

All three modes request 10 h / 7 d by default. Certificate lifetime and KDC policy constrain the initial grant.



Keep a job authenticated, then stop with it.

craft-maintain runs as the caller, coordinates one shared cache per UID, and watches a live process.

START ONCE / WATCH A SHELL THAT STAYS ALIVE FOR THE JOB

```
job_pid=$$
cache=$(/usr/local/bin/craft-maintain --watch-pid "$job_pid") || exit "$?"
export KRB5CCNAME="$cache"
```

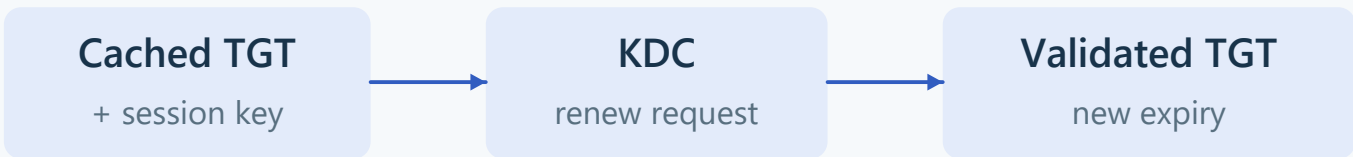
Returns when usable credentials and background maintenance are ready.

01 / RENEW

Use the current ticket

When a renewable TGT has $\leq 20\%$ of its lifetime left (capped at 2 h), and its expiry can extend.

A renewal error retries; no immediate fresh authentication.



Renew-till never moves later.
Retain usable service tickets.

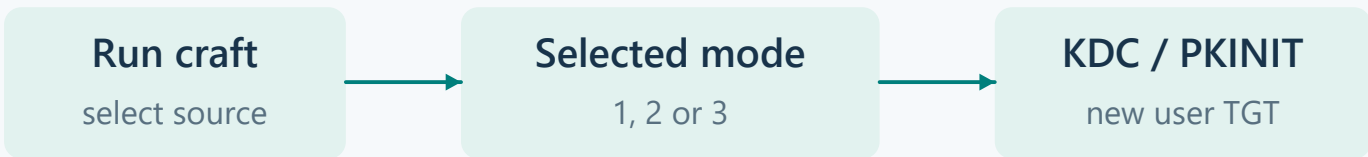
A due renewal takes priority during rollover, keeping usable credentials alive while fresh authentication is delayed.

02 / FRESH AUTHENTICATION

Obtain fresh credentials

Missing or expired TGT; renewal deadline is near; or a ticket that cannot extend is nearing expiry.

Rollover margin: $1/7$ of the renewal window, capped at 1 day.



Home PEM, enrollment or credential linking.
New renewal window; replace cache.

Successful refresh

Publish atomically to .krb5cc_craft (caller-owned, mode 0600).

Refresh failure

Keep the existing cache and retry; the job continues.

RETRY

Renewal: ~30 s to 5 min + jitter.
Fresh acquisition: 1 min to 1 h cooldown.

STOP

Watched PID exits, duration ends or signal.
Maintainer stops; the shared cache remains.

APPLICATION

Reload credentials when authenticating again.
Access follows the user's service permissions.